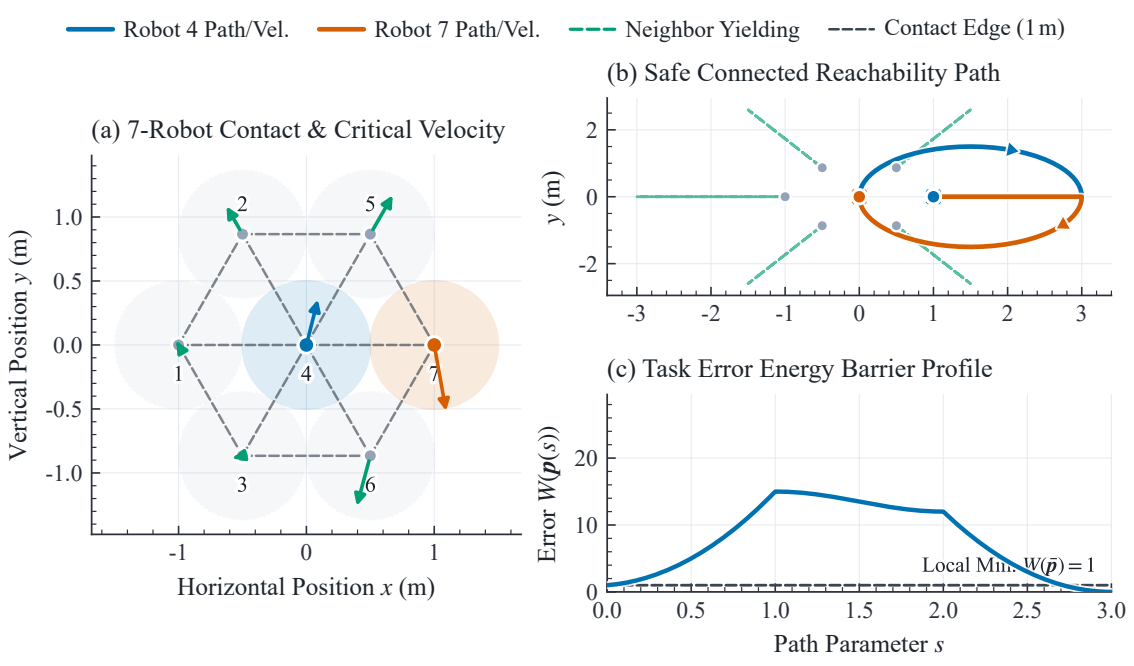

Robot 4 Path/Vel.
Robot 7 Path/Vel.
Neighbor Yielding
Contact Edge (1 m)
(a) 7-Robot Contact & Critical Velocity
Vertical Position y (m)
Horizontal Position x (m)
1
2
3
4
5
6
7
(b) Safe Connected Reachability Path
y (m)
(c) Task Error Energy Barrier Profile
Error W(p(s))
Local Min. W(p̄) = 1
Path Parameter s